AI-NATIVE ENGINEERING MASTERY

Transform Your Development Practice

Trainer: Pasi Vuorio | ModernPath

SESSION 1

AI-Assisted Development Fundamentals

90 min | Beginner

ModernPath

1. AI CODING TOOLS LANDSCAPE

Cursor

GitHub Copilot

Claude Code

Codex

RULES & CONFIGURATION

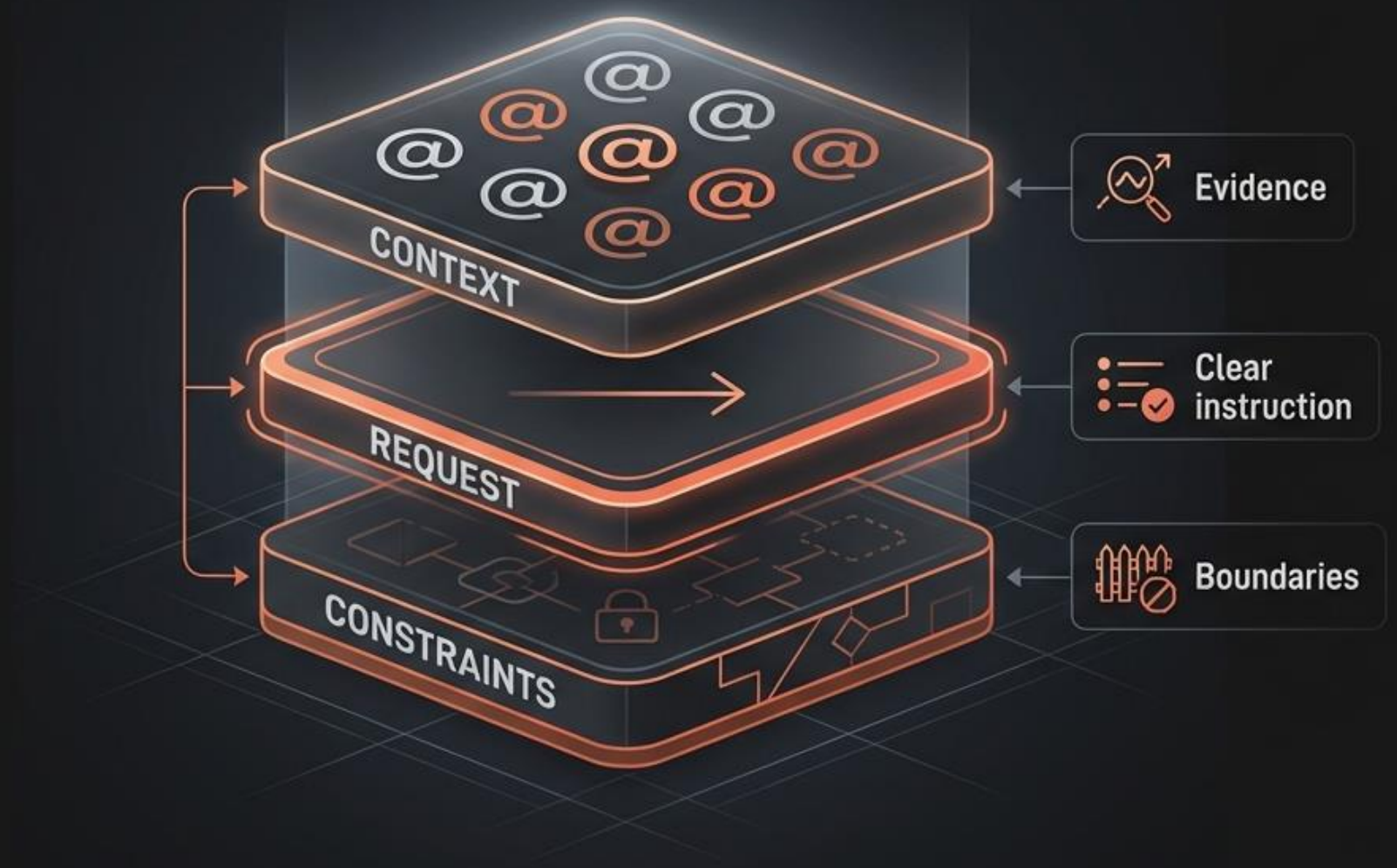
.cursor/rules/

CLAUDE.md

AGENTS.md

Your Codebase Constitution

2. EFFECTIVE PROMPTING PATTERNS



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THE SHIFT+TAB WORKFLOW

3. MODEL SELECTION

	CLAUDE	GPT	GEMINI
Reasoning stars:	★★★★☆	★★★★☆	★★★★★
Speed stars:	★★★☆☆	★★★★★	★★★☆☆
Context size:	200K tokens	128K tokens	1M+ tokens
Best For use case:	• Complex analysis • Nuanced writing • Ethical reasoning	• Creative content • Code generation • General knowledge	• Multimodal tasks • Large-scale retrieval • Real-time data integration

Match the brain to the task

4. GREENFIELD vs BROWNFIELD

GREENFIELD

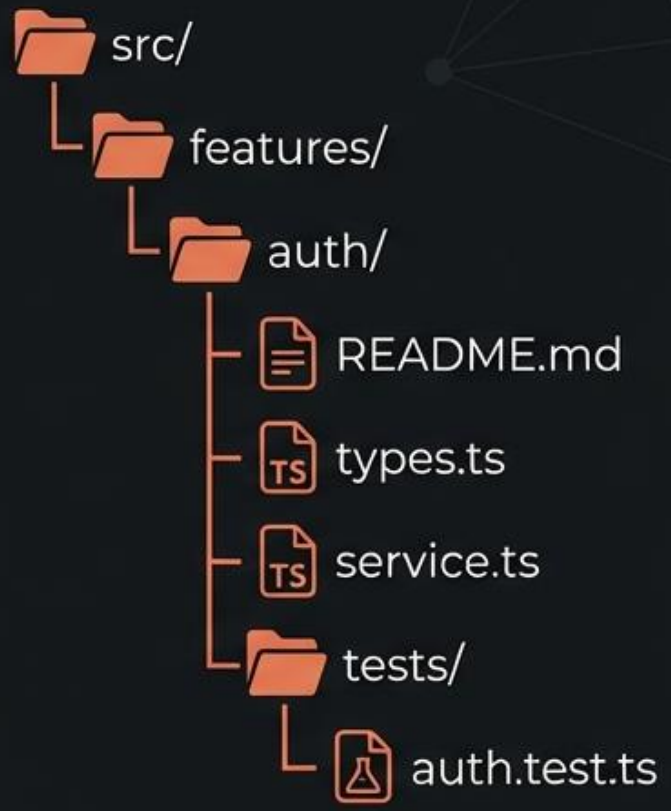
New Projects - Design for AI

BROWNFIELD

Existing Systems - Document First

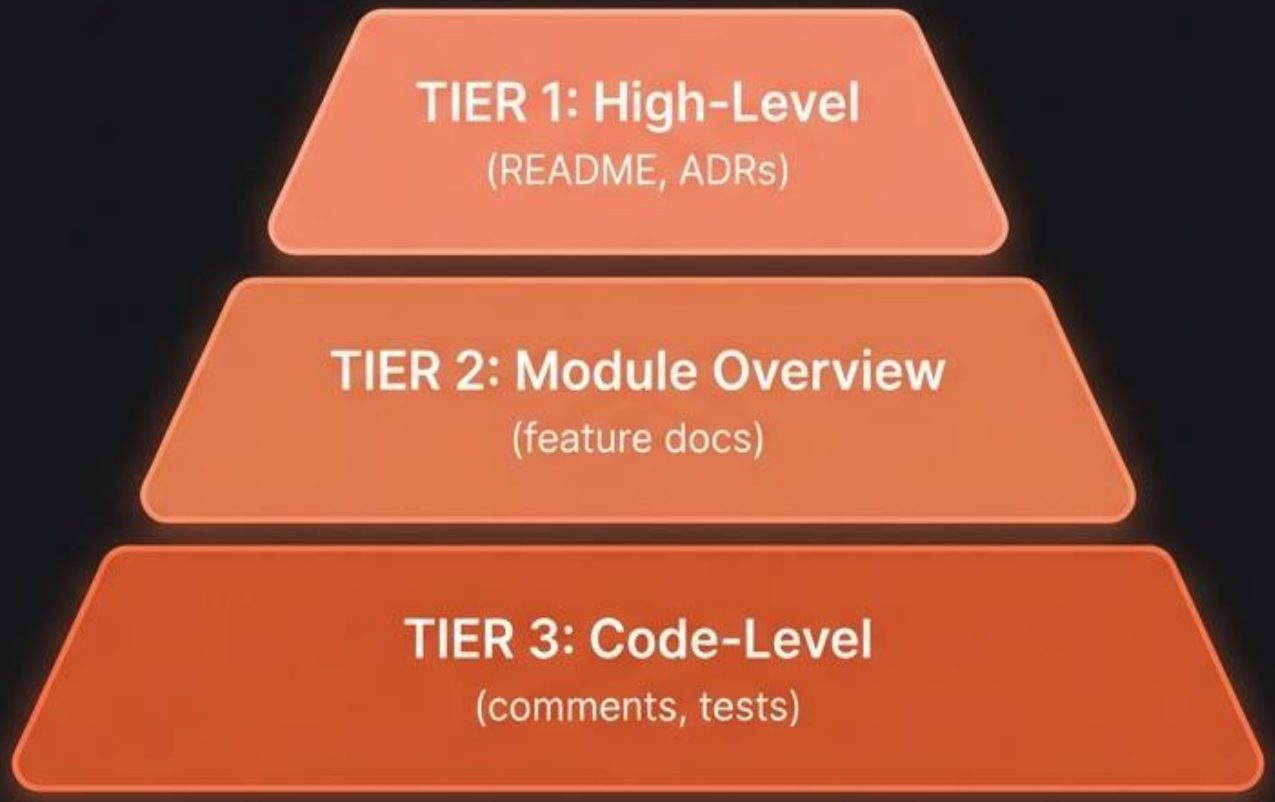
80% of enterprise work is brownfield

5. GREENFIELD: SPEC-DRIVEN DEVELOPMENT



Write the contract before the code

6. BROWNFIELD: 3-TIERED DOCUMENTATION



Your docs are the map for the AI brain

SESSION 2

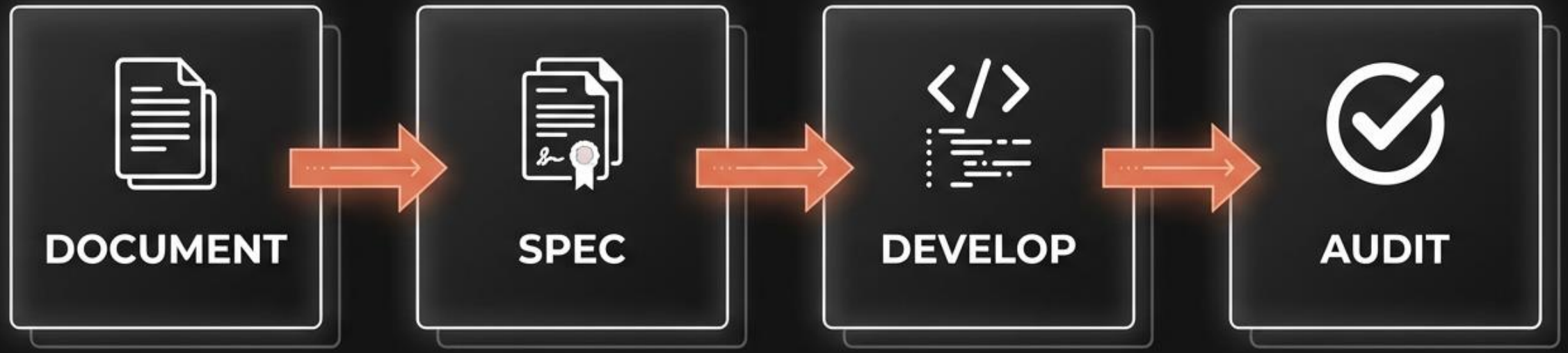
90 min | Intermediate

Spec-Driven Development & TDD with AI

ModernPath

MODERNPATH 4-STEP PROCESS

The Complete AI Development Lifecycle



WRITING EFFECTIVE SPECIFICATIONS

SPEC DOCUMENT TEMPLATE

- OVERVIEW
- USER STORIES
- ACCEPTANCE CRITERIA
- TECHNICAL CONSTRAINTS
- TEST STRATEGY

GIVEN-WHEN-THEN FORMAT EXAMPLE

Given a registered user is on the login page

And the user enters valid credentials

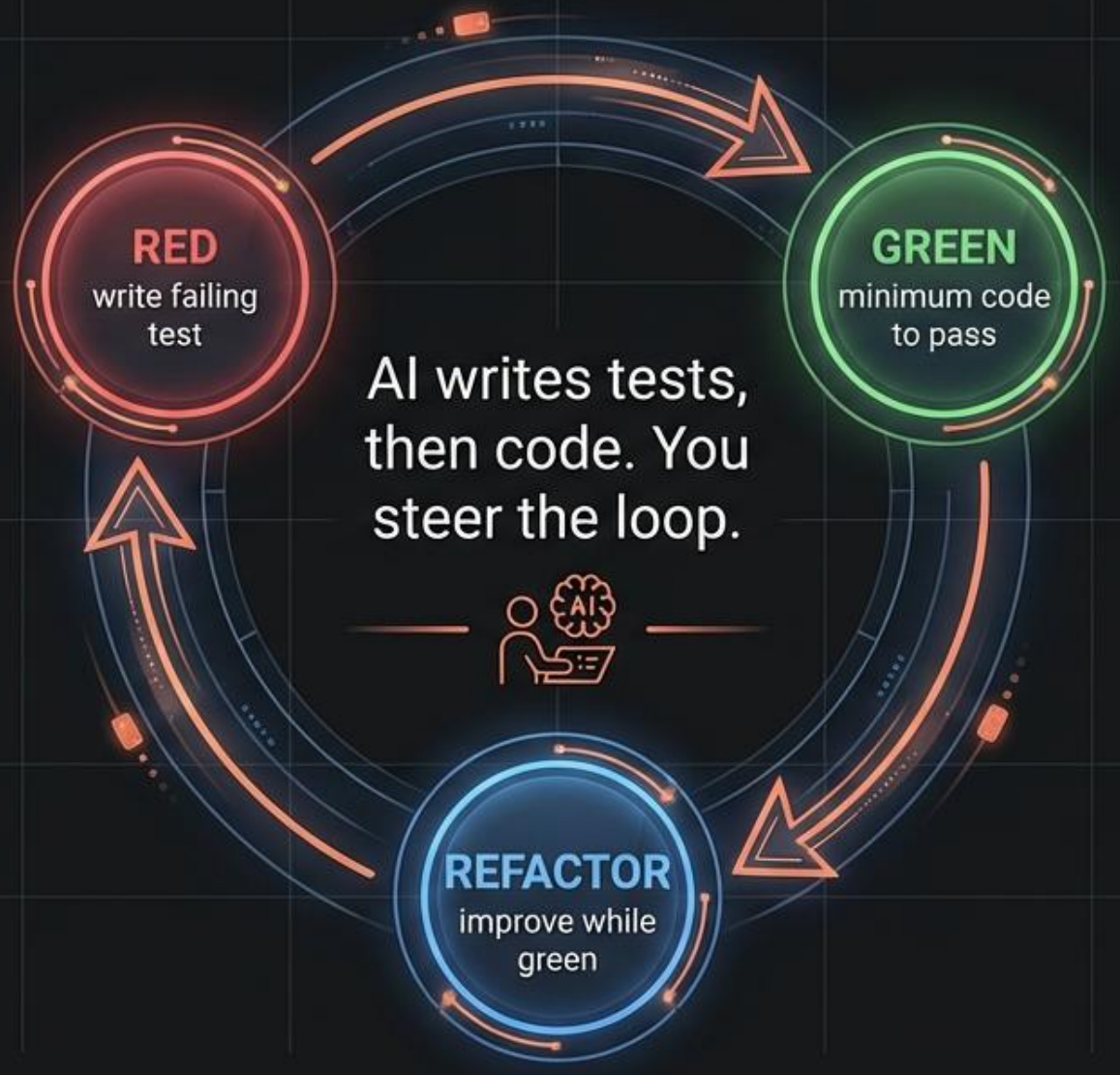
When the user clicks the "Login" button

Then the user is redirected to the dashboard

And a welcome message is displayed.

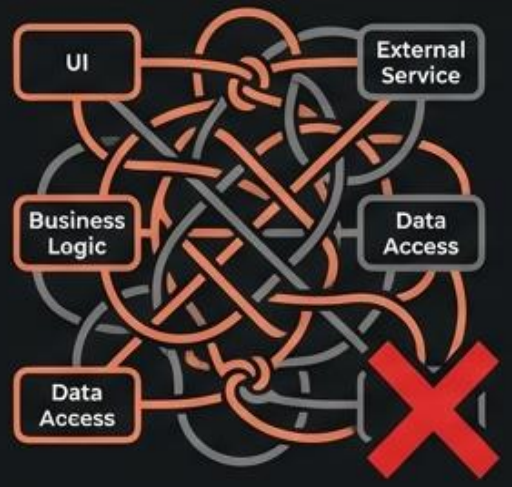
INSIGHT: IF AI CANNOT CONVERT YOUR AC TO A TEST, REWRITE IT

TDD CYCLE WITH AI



AI-FRIENDLY ARCHITECTURE

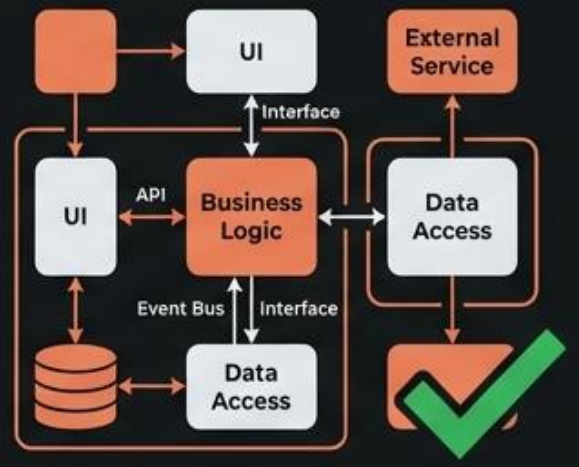
TIGHTLY COUPLED



- Changes cascade unpredictably.
- High dependencies create friction.
- High dependencies create friction.
- Difficult to isolate and test components.
- Risky for automated adjustments.

Problem: High Friction, Risky Edits

LOOSELY COUPLED



- AI modifies safely.
- Independent components with defined interfaces.
- Easy to replace or upgrade modules.
- Secure integration of AI capabilities.

Solution: Agility, Secure Integration

Depend on abstractions, not implementations.

SESSION 3

90 min | Advanced

Advanced Agentic Workflows

3

AGENTIC ARCHITECTURE

SPECIALIZED SUB-AGENTS

SPEC WRITER

Translates high-level requirements into detailed technical specifications, user stories, and acceptance criteria for the development team. Ensures clarity and completeness.

Focus: Requirements Engineering & Clarity

TEST GENERATOR

Automatically creates comprehensive test suites, including unit tests, integration tests, and end-to-end scenarios based on specifications. Enhances code coverage and quality.

Focus: Quality Assurance & Test Automation

CODE REVIEWER

Analyzes code submissions for best practices, potential bugs, security vulnerabilities, and adherence to coding standards. Provides actionable feedback.

Focus: Code Quality & Security Compliance

One agent cannot do it all. Build a team of specialists.

MCP vs CLI TOOLS

CLI	MCP
Universal	Structured
Text-based	Type-safe
Higher tokens	Lower tokens

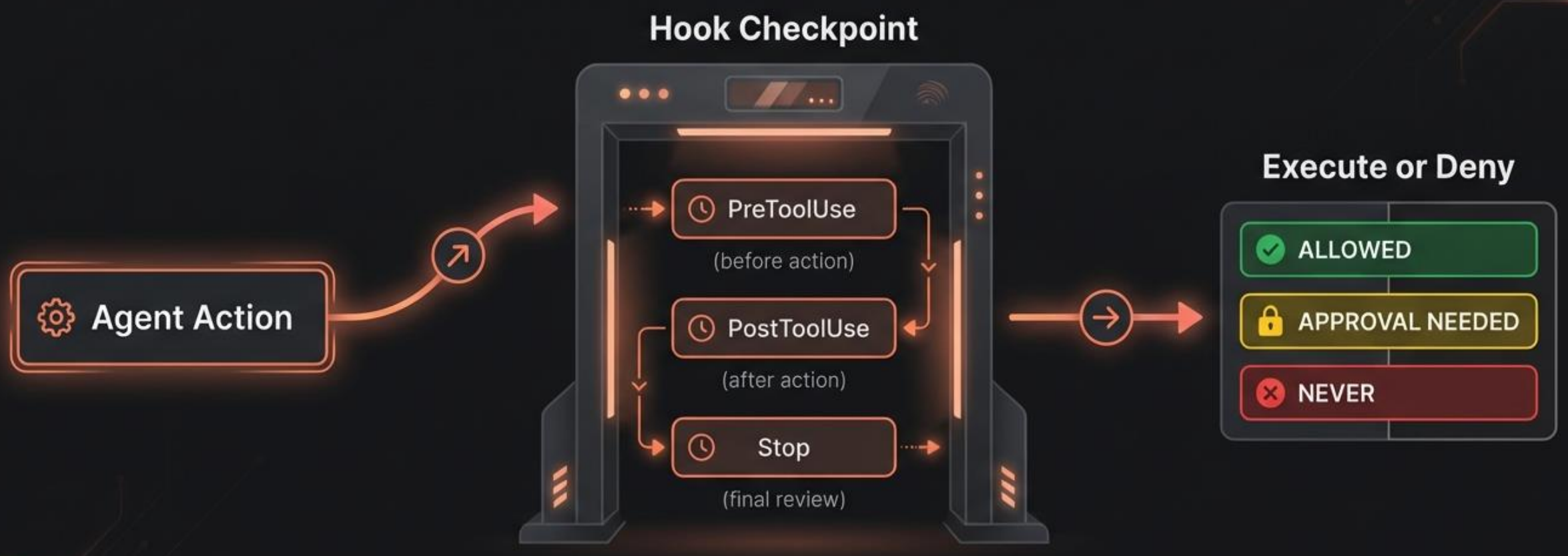
MCP for efficiency, CLI for debugging

Choose based on workflow needs and token economy.

MCP INTEGRATIONS

Connect AI to your company nervous system

HOOKS & VERIFICATION



Never let AI commit
without a green light

SCALING ACROSS TEAMS

METRICS

- Time to PR
- AI assist ratio
- First-pass success

Centralize standards, localize specifics